Type VI Secretion Apparatus in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review Brief

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBITaxon:160488; proteome UP000000556) **Target bucket:** KEGG ppu03070 — "Bacterial secretion system" (broad multi-system map; module area transport_motility_signaling) **Module under review:** Bacterial type VI secretion apparatus (contractile injection nanomachine) **Scope of curation task:** Manual module satisfiability and gene-annotation review — not a generic pathway essay.

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1. Executive Summary

The Type VI secretion (T6SS) apparatus module is **strongly satisfiable** in *Pseudomonas putida* KT2440, supported by **direct, species-specific experimental evidence**. The strain encodes **three complete or near-complete T6SS gene clusters** (designated K1, K2, K3) plus several orphan *hcp*/*vgrG*/PAAR islands that supply tube and spike machinery in trans. The **K1 cluster** (locus tags PP_3088–PP_3106, containing the sole genome-wide *clpV1* = PP_3095) is a **complete, standalone, experimentally validated antibacterial apparatus** encoding every one of the 13 canonical core subunits; it secretes Rhs-type effectors and kills competitor bacteria in the rhizosphere (Bernal et al. 2017, PMID: 28045455; Nie et al. 2022, PMID: 35178858).

Two curation caveats dominate this review. **First**, the KEGG bucket ppu03070 is a *broad, multi-system* "Bacterial secretion system" map mixing at least six export machineries (Sec/SRP, Tat, Type II/Xcp, Type I, Type V two-partner secretion, and Type VI). Its KEGG Orthology (KO) layer annotates only **6 of the 13 T6SS core families**; the entire sheath (TssB/TssC), the entire baseplate (TssE/F/G/K), and the TssA cap are **not KO-annotated** in this bucket. Module satisfiability therefore **cannot** be judged from KEGG KO calls and must be reconstructed from InterPro/Pfam domain evidence, UniProt, and primary literature. **Second**, this domain-level reconstruction exposes a concrete, curation-actionable **annotation error**: **PP_3099 is mislabeled as "uricase / urate oxidase (EC 1.7.3.3), gene *puuD*" when it is in fact the K1 sheath large subunit TssC1.** Correcting this single call converts the K1 cluster from apparently "missing TssC" to a complete 13/13 apparatus.

The residual genuine gaps are narrow and specific: **ClpV/TssH is single-copy** (ClpV1 = PP_3095, physically inside K1), so the sheath-recycling step for the K2 and K3 clusters must be trans-supplied or is cryptic; and the **K3 cluster lacks a detectable TssC** large sheath subunit (locus-tag gap at PP_4075). These are the only steps that should be flagged `candidate_uncertain`/`gap` for the K2/K3 sub-apparatuses. All other core steps are `covered` on domain evidence. Type III secretion is confirmed **absent** from KT2440 and should be marked `not_expected_in_target_taxon` where it appears in generic secretion overview maps.

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2. Target-Organism Pathway Definition

2.1 What the T6SS module is (and is not)

The T6SS is a **contractile bacterial secretion nanomachine** evolutionarily and structurally homologous to the tail of contractile bacteriophages (Brunet et al. 2015, PMID: 26460929). It comprises, mechanistically:

1. an **envelope-spanning membrane complex** (TssJLM) that docks the apparatus and provides the channel for the inner tube;
2. a **cytoplasmic baseplate** (TssEFGK plus VgrG) that serves as the tail-assembly platform and evolutionary adaptor;
3. a **contractile tail** — an **Hcp inner tube** wrapped by a **TssBC sheath**, coordinated by the **TssA cap**;
4. a **VgrG (+PAAR) puncturing tip**; and
5. the **ClpV (TssH) AAA+ ATPase** that recycles the contracted sheath.

Cargo effectors, cognate immunity proteins, effector-specific adaptors (Tap/DUF4123), and global transcriptional regulation lie **outside** the conserved structural boundary of this module and should be curated separately — though they are richly represented in KT2440 (10 effector–immunity pairs; Bernal 2017).

2.2 Pathway boundaries — what to keep separate

The KEGG bucket **ppu03070** is *not* a T6SS map; it is the pan-secretion overview for *P. putida* and must be decomposed. Programmatic KO classification of all 61 bucket genes shows it embeds at least six distinct systems:

Sub-system in ppu03070	Representative KO / genes	Curation action
Sec translocon / SRP	secA/Y/E/G/D/F, yajC, yidC, ffh, ftsY, secB (K03070–K03217)	Separate module (protein export)
Tat pathway	tatABC (K03116–K03118)	Separate module
Type II secretion (Xcp/Gsp)	gspC–L (K02452–K02461)	Separate module
Type I secretion	hlyB/hlyD (K11003/K11004), tolC (K12340)	Separate module
Type V two-partner secretion	shlA/B (K11016/K11017)	Separate module
Type VI secretion	**K11891/K11892/K11903/K11904/K11906/K11907**	**This module**

Critically, the **truncated candidate list surfaced in the commissioning brief** (yidC, sec*, tat*, xcp*/gsp*) is dominated by Sec/Tat/T2SS genes and **does not represent the T6SS at all**; the actual T6SS genes are among the 45 candidates omitted from the prompt. Any reviewer working only from the visible candidate names would mis-scope the module. **Type III secretion is absent** from KT2440 (no *hrc*/*sct* secretion apparatus; apparent *sctC*/*sctS* hits resolve to sulfur-compound ABC transporters), so it should not be expected in this organism.

2.3 Alternate names / database definitions

Core subunits carry multiple synonymous names that fragment free-text annotation: TssB≡VipA≡EvpA; TssC≡VipB≡EvpB; TssH≡ClpV; TssD≡Hcp; TssI≡VgrG; TssM≡IcmF≡ImpL≡VasK; TssL≡DotU≡VasF; TssJ≡VasD/Lip. KEGG KO IDs used in ppu03070: K11891 (TssM), K11892 (TssL), K11906 (TssJ/VasD), K11903 (Hcp), K11904 (VgrG), K11907 (ClpV). There is no dedicated stand-alone "T6SS" KEGG map. KT2440's clusters additionally use a **system-specific numeral** ("1") convention: **ClpV1** / **TssB1** / **TssC1** all belong to the K1 locus — a useful internal consistency check.

3. Expected Step Model and Satisfiability

The following table maps each canonical step to KT2440 evidence, per cluster, and gives the recommended module curation status.

Module step	Core player	KT2440 genes (K1 / K2 / K3)	KO in ppu03070?	Curation call
1. Membrane complex	TssJ (lipoprotein)	PP_3094 / PP_2618 / PP_4079	✔ K11906	**covered**
	TssL (DotU)	PP_3092 / PP_2616 / PP_4081 (+orphan PP_3385)	✔ K11892	**covered**
	TssM (IcmF)	PP_3090+PP_3091 / PP_2627 / PP_4071	✔ K11891	**covered** (K1 TssM split across 2 ORFs)
2. Baseplate	TssE (Gp25-like)	PP_3098 / PP_2622 / PP_4076	✖	**covered** (UniProt/ Pfam), KEGG gap
	TssF	PP_3097 / PP_2621 / PP_4077	✖	**covered**, KEGG gap
	TssG	PP_3096 / PP_2620 / PP_4078	✖	**covered**, KEGG gap
	TssK	PP_3093 / PP_2617 / PP_4080	✖	**covered**, KEGG gap
3. Tube + sheath	TssA (cap)	PP_3088 / PP_2626 / PP_4072	✖	**covered** (ImpA N-term), KEGG gap
	Hcp (tube)	PP_3089 / PP_2615 / PP_4082 (+orphans PP_4886, PP_5238, PP_0655)	✔ K11903	**covered**
	TssB (sheath small)	**PP_3100 (IPR008312=TssB1)** / PP_2624 / PP_4074	✖	**covered** all three (K1 only via InterPro)
	TssC (sheath large, VipB)	**PP_3099 (mis-annot. "uricase"→TssC1)** / PP_2623 / **absent (PP_4075 gap)**	✖	**covered** K1 & K2; **gap** K3
	VgrG		✔ K11904	**covered**

Module step	Core player	KT2440 genes (K1 / K2 / K3)	KO in ppu03070?	Curation call
4. Puncturing tip		PP_3106 / PP_2614 / PP_4049 (+orphan vgrG-II PP_3386)		
	(PAAR sharpener)	— / PP_2610 / PP_4045	✖	accessory; covered where present
5. Sheath recycling	ClpV (TssH)	**PP_3095 (K1 only; sole genome-wide ClpV)**	✔ K11907 (single)	**covered K1**; **candidate_uncertain K2/K3**

Genome-wide there is exactly one T6SS ClpV (PP_3095); PP_0625/PP_4008/PP_3316 are housekeeping ClpB/ClpA-type chaperones, not T6SS. Type III secretion is absent in KT2440.

Verified per-cluster completeness (Pfam-based audit):

Cluster	Core subunits present	ClpV	Verdict
K1 (PP_3088–3106, *clpV1*)	**13/13** (incl. TssB1 PP_3100, TssC1 PP_3099)	✔ PP_3095	**complete standalone apparatus** (= functional K1-T6SS)
K2 (PP_2610–2628, *vgrG-I*)	12/13 (full TssBC sheath, baseplate, membrane)	✖	complete except ClpV → recycling trans-shared
K3 (PP_4045–4049 + PP_4071–4085, *vgrG-III*)	11/13 (**no TssC, no ClpV**)	✖	cryptic/trans-dependent

TssC (Pfam PF05943/PF18945) occurs in exactly two proteins genome-wide (PP_2623 K2, PP_3099 K1); the sole T6SS ClpV is PP_3095. TssJ/TssL confirmed by curated name; all other subunits confirmed one-per-cluster by discriminating Pfam.

4. Candidate Genes and Evidence

4.1 Finding F001 — Three T6SS clusters plus orphan islands (direct, target-strain evidence)

P. putida KT2440 was genome-analyzed by Bernal et al. (2017), who identified three T6SS gene clusters (K1-, K2-, K3-T6SS) and 10 effector–immunity pairs. The K1 system is a **potent antibacterial device** secreting the Rhs effector Tke2, active against a broad range of bacteria in the rhizosphere. This is **direct experimental evidence in the target strain** and is the strongest possible support for module presence.

> "Here we analyze the genome of the biocontrol agent *Pseudomonas putida* KT2440 and identify three T6SS gene clusters (K1-, K2- and K3-T6SS). Besides, 10 T6SS effector-immunity pairs were found" — PMID: 28045455
>
> "*Pseudomonas putida* KT2440 possesses a functional T6SS (K1-T6SS) and exhibits antibacterial activity towards a broad range of bacteria" — Nie et al. 2022, PMID: 35178858

Subsequent work confirms functional, engineerable T6SSs in this exact strain: KT2440 has been engineered to heterologously deliver antibacterial and antifungal effectors via its T6SS (PMID: 40176102); the three systems shape rhizosphere and biofilm community structure (PMID: 41036489; PMID: 38096690); the systems are transcriptionally distinct and independently regulated (PMID: 36748579); and a cryo-EM structure of the KT2440 Tap3–Tke5 adaptor–effector complex has been solved (PMID: 41526723). The module is unambiguously real and active in the target organism.

4.2 Finding F002 — KEGG KO layer covers only 6 of 13 core families

Programmatic KEGG REST mapping of all 61 ppu03070 genes to KO IDs shows the T6SS panel is KO-annotated for only six families: **TssM/IcmF** (K11891; PP_2627, PP_3090, PP_3091, PP_4071), **TssL/DotU** (K11892; PP_2616, PP_3092, PP_3385, PP_4081), **TssJ** lipoprotein (K11906; PP_2618, PP_3094, PP_4079), **Hcp** (K11903; PP_2615, PP_3089, PP_4082, PP_4886, PP_5238), **VgrG** (K11904; PP_2614, PP_3106, PP_3386, PP_4049), and **ClpV** (K11907; single copy PP_3095). There is **no KO annotation in this bucket** for the sheath (TssB, TssC), the baseplate (TssE/F/G/K), or the TssA cap. Since the canonical apparatus comprises 13 core subunits (Brunet 2015, PMID: 26460929), the KO layer leaves **7 of 13 steps invisible**. KEGG gene descriptions here are largely "conserved protein of unknown function," signalling shallow annotation depth. **Curation consequence:** do not judge module satisfiability from KEGG KO coverage — it will produce false gaps.

> "The 13 T6SS subunits assemble a cytoplasmic contractile structure anchored to the cell envelope by a membrane-spanning complex." — PMID: 26460929

4.3 Findings F004/F005 — Full per-cluster reconstruction from UniProt; ClpV single-copy

Reconstruction of the three clusters from UniProt proteome UP000000556 resolves **every canonical core subunit**, most unannotated in KEGG (Section 3 table). Key facts for curation:

- **TssM/IcmF** in K1 is **split across two ORFs** (PP_3090 + PP_3091) — a real biological/annotation feature to preserve, not merge blindly.
- **ClpV/TssH is single-copy genome-wide**: a proteome-wide search for ClpV/ClpB AAA+ ATPases returns exactly **one** T6SS ClpV, **ClpV1** = **PP_3095**, embedded in K1. The only other Clp ATPases are housekeeping chaperones (ClpB = PP_0625, ClpA = PP_4008, PP_3316), **not** T6SS-associated. The K2 (PP_2610–2628) and K3 (PP_4071–4085) windows contain **no** *clpV*.
- **Sheath annotation is asymmetric on free-text names**: K2 has clearly named TssB (PP_2624, 167 aa) and TssC (PP_2623, 496 aa); K1's sheath was initially represented only by PP_3100 ("Type VI secretion protein," 191 aa) with the operon apparently interrupted by an unrelated "uricase" gene; K3 has TssB (PP_4074) but its expected large-subunit locus (PP_4075) is unannotated.

High-confidence, direct target-organism support exists for the Hcp tube (secretion is the classic functional readout, demonstrated for K1), the VgrG tip (engineered VgrG-dependent delivery shown in KT2440), ClpV1, and the three TssJLM membrane complexes. Baseplate TssE/F/G/K, sheath TssB/TssC, and TssA are **same-organism UniProt/Pfam** calls in canonical operon order — transfer is **strong** because these are same-genome ORFs, not cross-species inference.

4.4 Finding F006 — InterPro rescues subunits that free-text names miss

InterPro/Pfam annotation resolves what free-text names hide. **PP_3100** (191 aa), named merely "Type VI secretion protein," carries **IPR008312 = T6SS sheath protein TssB1** — it *is* the K1 small sheath subunit, and the shared "1" (ClpV1/TssB1) confirms a coherent K1 locus. Baseplate and membrane families are cleanly resolved in all clusters by InterPro (TssK IPR010263, TssF

IPR010272, TssG IPR010732, ClpV1 IPR017729). **Curation consequence:** InterPro/Pfam is the authoritative layer for this module in KT2440; name-based searches systematically under-count.

4.5 Finding F007 — PP_3099 "uricase" is actually TssC1 (a real annotation error)

This is the single most curation-actionable finding. **PP_3099** (500 aa) is annotated in UniProt as *"Uricase/urate oxidase (EC 1.7.3.3), gene name puuD"*, yet domain analysis shows it carries **IPR010269 (TssC-like), IPR044031 (TssC1 N-terminal), IPR044032 (TssC1 C-terminal), and Pfam PF05943 + PF18945 (EvpB/VipB tail-sheath domains)** — identical to the bona fide TssC PP_2623 (K2; same Pfams) — and carries **no uricase Pfam (PF01014)**. Its **genomic position is exactly the expected TssC slot**, between TssE (PP_3098) and TssB1 (PP_3100), inside the K1/ClpV1 operon. A genome-wide scan for the sheath-large domain (PF05943/PF18945) returns **exactly two** proteins: PP_2623 (K2) and PP_3099 (K1). The complementary TssB scan (IPR008312/PF05591) returns **exactly three** (PP_2624 K2, PP_3100 K1, PP_4074 K3) — one per cluster.

Correcting PP_3099 → TssC1 completes the K1 cluster as a 13/13 standalone apparatus, consistent with its experimentally validated antibacterial function. A curation trap to note: **IPR010269 ("TssC-like") is a broad superfamily also hit by TssF** (PP_2621, PP_4077); discriminate TssC from TssF using PF05943 (TssC) vs PF05947 (TssF).

4.6 Finding F008 — Verified per-cluster completeness matrix

Systematic Pfam-based enumeration binned to clusters confirms **one copy per cluster** for TssA (PF06812), TssB (PF05591), TssE (PF04965), TssF (PF05947), TssG (PF06996), TssK (PF05936), TssM (PF06761), Hcp (PF05638; plus orphans PP_4886/PP_5238/PP_0655), and VgrG (PF05954; plus orphan vgrG-II PP_3386). TssJ and TssL are confirmed by curated names. **TssC (PF05943) returns exactly two proteins genome-wide** — PP_3099 (K1) and PP_2623 (K2) — so **K3 has no TssC**. **Net: K1 = 13/13, K2 = 12/13 (no ClpV), K3 = 11/13 (no TssC, no ClpV).**

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5. Mechanistic Model / Interpretation

5.1 Assembly logic (supported by structural literature)

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OUTER MEMBRANE ┌── TssJ (OM lipoprotein) ──────────────┐
               │                                       │  MEMBRANE COMPLEX
INNER MEMBRANE ├── TssL (DotU) ── TssM (IcmF) ─────────┘  (TssJLM)
               │         │  recruits baseplate via TssG/TssK contacts
CYTOPLASM      ▼         ▼
          TssE·TssF·TssG·TssK + VgrG ── BASEPLATE (assembly platform)
                        │  TssA caps and coordinates polymerization
                        ▼
          Hcp inner tube  ⇄  TssB/TssC sheath (extended)
                        │  sheath contracts → propels VgrG(+PAAR) tip + effectors
                        ▼
          ClpV (TssH) AAA+ ATPase recycles contracted sheath → next firing
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This assembly order — membrane complex → baseplate → tail tube/sheath → contraction → ClpV recycling — is established by structural and biochemical work in enteroaggregative *E. coli* (EAEC): baseplate biogenesis and structure (PMID: 30323254), TssEFGK–VgrG recruitment to TssJLM (PMID: 26460929), TssL–baseplate–membrane contacts (PMID: 27600409), the TssM–TssG interface (PMID: 27600411), VgrG–Hcp adaptor function (PMID: 30031895), TssB sheath structure/function (PMID: 29223729), and the TssA cap (PMID: 28817192). Transfer of this framework from EAEC to KT2440 is **strong** at the level of subunit identity and assembly order (universally conserved architecture), but fine details (e.g., the domesticated MltE transglycosylase required for membrane-complex assembly through peptidoglycan, PMID: 27920034) are **inferred, not directly demonstrated in KT2440.**

5.2 The three-cluster division of labor in KT2440

The most parsimonious interpretation is that **K1 is a fully autonomous apparatus** (13/13, with its own ClpV1 and validated antibacterial output), whereas **K2 and K3 are structurally near-complete tails that depend on trans-acting or shared components** for two steps:

- **Sheath recycling (ClpV):** With only one ClpV in the genome (ClpV1/PP_3095, physically in K1), the K2 and K3 sheaths must either (a) be recycled by ClpV1 acting in trans, (b) fire only once without recycling, or (c) be recycled by an as-yet-unrecognized enzyme. ClpV recognizes a cognate TssC N-terminal helix, so cross-recycling is not guaranteed. This step is genuinely **candidate_uncertain** for K2/K3.

- **TssC in K3:** The K3 locus has TssB (PP_4074) but no detectable TssC (PF05943) — the slot PP_4075 is not present as a separate VipB-domain ORF. K3 may therefore be **degenerate/ pseudogenized**, or its large sheath subunit may be supplied in trans by TssC1 or TssC2. This is the one genuine core **gap**.

The three systems are transcriptionally distinct and independently regulated (PMID: 36748579), consistent with functional specialization rather than pure redundancy.

5.3 Why the KEGG bucket misleads

The confusion at every level of this review traces to two annotation-layer facts: (1) ppu03070 is a **pan-secretion overview** intermixing six systems, and (2) its KO layer covers fewer than half the T6SS core families and leaves the sheath/baseplate/cap as "unknown function." The correct KT2440 T6SS annotation is only visible at the **InterPro/Pfam** layer — which additionally corrects a false EC-numbered enzyme call (PP_3099 "uricase"). This is a textbook example of why module satisfiability must be judged on domain evidence and primary literature, not on pathway-database bucket membership.

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6. Gaps, Ambiguities, and Likely Over-Annotations

Item	Type	Locus	Recommendation
PP_3099 "Uricase (EC 1.7.3.3), *puuD*"	**Annotation error**	K1	Re-annotate as **TssC1** (sheath large subunit); remove EC 1.7.3.3; promote to `fetch-gene`
ClpV single-copy	Real functional gap for K2/K3	PP_3095 (K1 only)	Mark K2/K3 recycling step `candidate_uncertain`; test trans-recycling
K3 TssC absent	Real core gap	slot PP_4075	Mark K3 sheath-large step `gap`; check for pseudogene / trans-sharing
PP_3100 "Type VI secretion protein"	Under-annotation	K1	Promote to **TssB1** (InterPro IPR008312)
IPR010269 "TssC-like" hitting TssF	**Over-broad domain**	PP_2621, PP_4077	Do not assign TssC from IPR010269 alone; require PF05943/PF18945
K11892 TssL/DotU multi-mapping	Paralog ambiguity	PP_2616, PP_3092, PP_3385, PP_4081	Resolve per-cluster; PP_3385 is an orphan tssL near orphan vgrG PP_3386
Hcp/VgrG multi-copy	Expected biology, not error	orphans PP_4886/ PP_5238/PP_0655, PP_3386	Curate as trans-acting tube/spike modules or effectors, not duplicates
TssM split ORF in K1	Structural feature	PP_3090+PP_3091	Preserve as two ORFs; do not force-merge
Type III secretion	Absent	—	Mark `not_expected_in_target_taxon` in generic maps
Sec/Tat/SRP/ T1SS/T2SS/ T5SS in ppu03070	Boundary over-inclusion	many	Do not pull into the T6SS module; KEGG-map artifact

Likely over-propagation risk: the broad "TssC-like" superfamily (IPR010269) and the generic KO "unknown function" labels are the two places most likely to generate false or missing T6SS annotations for related *Pseudomonas* proteomes. Any automated pipeline that trusts EC 1.7.3.3 on PP_3099 will silently break the K1 module. **Cluster-naming caveat:** the K1/K2/K3 labels here are mapped to genomic loci by inference (the ClpV1-containing locus PP_3088–3106 → "K1"); the exact K2 vs K3 correspondence should be confirmed against Bernal et al. before hard-coding into module documents.

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7. Module and GO-Curation Recommendations

- **Mark covered:** membrane complex (TssJLM), baseplate (TssEFGK), Hcp tube, sheath (TssBC), VgrG tip, and ClpV recycling are ALL covered for the **K1 cluster** (a complete standalone apparatus once PP_3099 is corrected to TssC1). K2 is covered for all steps except ClpV; at the genome level every core family is present.
- **Mark candidate_uncertain / gap:** ClpV-dependent recycling for K2/K3 (single genome-wide ClpV1); TssC for K3 (no PF05943/PF18945 protein at slot PP_4075).
- **Correct an annotation error:** PP_3099 must be re-typed from uricase (EC 1.7.3.3) to TssC1 — the key data-quality action.
- **Not a gap / not_expected:** none of the core steps should be marked "not expected"; the full core is present. Type III secretion, a *neighboring* system, *is* `not_expected_in_target_taxon`.
- **module_needs_revision (generic boundary):** the generic module treats the apparatus as single-copy. For KT2440 the module should be **instantiated three times (K1/K2/K3)** or carry an explicit "paralogous multi-copy" flag, and the single-ClpV topology (one ATPase potentially serving multiple clusters) should be representable. The architectural boundary itself (membrane complex → baseplate → tail → tip → recycling) is **correct** and does not need revision.
- **Annotation-source note:** because KEGG ppu03070 KO-annotates only 6/13 core families, satisfiability must be evaluated against **UniProt/InterPro/Pfam**, not KEGG KO alone. Recommend adding InterPro signatures (TssA/ImpA, TssE/Gp25-like, TssF/TssG, TssK, VipA/VipB sheath) to the module's evidence sources.
- **GO usage:** structural subunits map to *T6SS complex* (GO:0033104) and *protein secretion by the T6SS* (GO:0033103); ClpV to ATP hydrolysis activity (GO:0016887). No new GO term appears strictly required, but a GO/annotation request to capture per-cluster sheath (TssB/TssC) and TssA in KEGG would close the KO gap.

8. Genes to Promote to Full `fetch-gene` Review

1. **PP_3099 — top priority: correct "uricase (EC 1.7.3.3)" → TssC1** (sheath large subunit; PF05943/PF18945). A clear erroneous annotation that also completes the K1 sheath.
2. **PP_3095 (ClpV1)** — sole genome-wide recycling ATPase; test whether it services K2/K3 in trans (defines their functionality).
3. **PP_4075 slot / K3 TssC** — confirm true loss (pseudogene, frameshift, or genuine absence) before finalizing the K3 `gap`.
4. **PP_3100 (TssB1, currently "Type VI secretion protein")** — rename to TssB (IPR008312).
5. **PP_3090 + PP_3091 (K1 TssM/IcmF)** — resolve whether this is a split/fused TssM annotation, not an assembly artifact.
6. **PP_3385 / PP_3386 (orphan tssL / vgrG-II)** — clarify whether these form a fourth partial system or are trans-acting accessories.
7. **Orphan Hcp (PP_4886, PP_5238, PP_0655)** — determine whether tube components of a cluster or standalone effector/immunity modules.

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9. Limitations and Knowledge Gaps

- **Cluster-to-window binning** relied on locus-tag adjacency; a few accessory genes at cluster edges may be mis-binned. This does not affect the core-subunit calls, which are domain-anchored.
- **ClpV trans-recycling for K2/K3 is a hypothesis**, not demonstrated. Whether ClpV1 recognizes K2/K3 TssC N-termini is untested in KT2440.
- **K3 TssC absence** is a negative result from domain scanning; a highly divergent or frameshifted TssC could evade PF05943 detection. This warrants direct sequence inspection of the PP_4074–PP_4076 interval.
- **Mechanistic transfer from EAEC** (assembly order, MltE peptidoglycan domestication, VgrG–Hcp adaptor mechanism) is homology-based; target-strain evidence is strongest for K1 *function* (Bernal 2017) but not for step-by-step *assembly* in KT2440.

- This review deliberately **excluded** effectors, immunity proteins, adaptors, and regulation from the module boundary; KT2440's rich effector repertoire (10 E–I pairs; Tke2, Tke5, Tap3) is documented but out of module scope.

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10. Proposed Follow-up Experiments / Actions

1. **Re-annotate PP_3099 → TssC1** in the local metadata and submit an error report to UniProt/KEGG (removes spurious EC 1.7.3.3).
2. **Sequence-inspect the K3 PP_4074–PP_4076 interval** for a cryptic/frameshifted TssC before finalizing the K3 `gap` call.
3. **Test ClpV1 trans-function:** a Δ*clpV1* mutant assayed for K2- and K3-dependent secretion would resolve whether recycling is shared or K1-exclusive.
4. **Confirm the K1 TssM split** (PP_3090/PP_3091) by RNA-seq operon structure or targeted sequencing.
5. **Run an InterPro/Pfam sweep as the primary annotation layer** for all *Pseudomonas* T6SS curation, using PF05943 (TssC) vs PF05947 (TssF) to avoid the IPR010269 superfamily trap.
6. **Expert question for T6SS specialists:** Do the three KT2440 systems share the single ClpV1, and is K3 a functional or degenerate apparatus?

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11. Key References

PMID	Relevance & evidence type
28045455	**Primary target-strain evidence:** three T6SS clusters (K1/K2/K3) and 10 effector–immunity pairs in KT2440; K1 antibacterial function (Tke2). *Direct.*
35178858	Confirms K1-T6SS is a functional antibacterial apparatus in KT2440; c-di-GMP/Wsp regulation. *Direct.*
36748579	Transcriptional organization/regulation of the three *P. putida* T6SSs. *Direct.*
40176102	Engineering KT2440 T6SS for VgrG/PAAR-dependent effector delivery. *Direct.*
41036489; 38096690	KT2440 T6SS shapes rhizosphere/biofilm community structure. *Direct.*
41526723	Cryo-EM of KT2440 Tap3–Tke5 adaptor–effector complex (effector biology; outside structural module). *Direct.*
26460929	Canonical 13-subunit core; TssEFGK–VgrG baseplate recruitment to TssJLM — the reference model for satisfiability. *Generic; strong transfer.*
30323254	Cryo-EM baseplate structure and biogenesis pathway (EAEC). *Generic; structural reference.*
27600409	TssL–baseplate–membrane complex interactions. *Generic.*
27600411	TssM cytoplasmic domain–TssG baseplate interface. *Generic.*
30031895	VgrG serves as adaptor to nucleate Hcp tube assembly. *Generic.*
29223729	TssB sheath subunit structure/function. *Generic.*
28817192	TssA cap coordinates tube/sheath polymerization. *Generic.*
27920034	MltE transglycosylase domestication for membrane-complex assembly (mechanistic detail, EAEC). *Generic; inferred for KT2440.*

Evidence-strength summary

- **Direct experimental (KT2440):** presence of three clusters; K1 functional antibacterial activity; Hcp/effector secretion; regulation by c-di-GMP/Wsp; engineered effector delivery; Tap3–Tke5 structure.
- **Same-organism sequence/UniProt annotation (strong):** completeness of all 13 core subunits in K1; near-completeness of K2/K3.
- **Cross-species homology (conserved core, strong transfer):** subunit roles/architecture and assembly order from EAEC/*E. coli* structural work.
- **Open questions:** ClpV provisioning for K2/K3; firing status of K2/K3; true absence vs. cryptic K3 TssC; whether orphan *hcp*/*vgrG* are standalone effectors.

Report prepared for manual module satisfiability and gene-annotation curation. Bottom line: module presence in KT2440 is direct and experimentally validated (K1); per-step completeness is domain-evidence-based (InterPro/Pfam); the mislabeling of PP_3099 as uricase and the single-copy ClpV / K3-TssC questions are the concrete curation actions.

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